MAT1072 - MATHEMATICS 2 MIDTERM EXAM — D

Name Surname:

Student ID:

Department/Group:

Answer Key

Duration: 100 mins

Date: 11.04.2026

Signature:

! Attention: *According to Article 54 of the Higher Education Law No. 2547 and Article 7/e of the Higher Education Institutions Student Disciplinary Regulati prepared pursuant to this article, those who commit the acts described as "Cheating or assisting in cheating during exams," and under Article 8/d as "Cheati by means of threats during exams, preventing the removal of students who are cheating from the examination hall, having someone else take the exam in one place, or taking the exam in place of someone else," will be subject to a suspension penalty of one or two semesters in the disciplinary investigation to be initiate*

1. Find the limit of the following function, if exists.

$$\lim_{(x,y)\to(0,0)} \frac{y(1-\cos x)}{x^4+y^2}$$

a) Does not exist b) 1 c) 0 d) -1 e) 2

IMPORTANT NOTE: The exam grade of those who do not mark the booklet type on the optical answer sheet will be entered as **ZERO** into the OBS system.

2. Which of the following correctly gives the critical points and their classification for the function

$$f(x,y) = x^3 - y^2 + 6xy + 3x + 4y \ ?$$

a) $(-1,-1)$ is a saddle point and $(-5,-13)$ is a local minimum point.

b) $(-1,-1)$ is a saddle point and $(-5,-13)$ is a local maximum point.

c) $(1,1)$ is a saddle point and $(-5,-13)$ is a local maximum point.

d) $(-5,-13)$ is a local minimum point and $(1,1)$ is a saddle point.

e) $(1,1)$ and $(-5,-13)$ are both saddle points.

3. Which of the following is the approximate value of the function $f(x,y) = e^{x^2-y}$ at the point $(1.02, 0.98)$, calculated using its linearization at the point $(1,1)$?

a) 1.08 b) 1.02 c) 1.06 d) 1.04 e) 1.10

4. Let b be a real number and the function

$$f(x,y) = e^{x+y}\left(\cos(xy) - b\ln(x^2+y^2)\right)$$

is given.

If

$$f_x(1,0) = 1$$

which of the following is the constant b?

a) $\frac{e+1}{2e}$ b) $\frac{1-e}{2e}$ c) $\frac{e-1}{e}$ d) $\frac{e-1}{2e}$ e) $\frac{1}{2e}$

5. Which of the following options gives the double integral for the volume of the solid formed inside the cylinder $x^2+y^2=9$, below the paraboloid $z = 9-x^2-y^2$, and above the plane $z=0$?

a) $\int_{-3}^{3}\int_{-3}^{3}(9-x^2-y^2)\,dy\,dx$

b) $\int_{-3}^{3}\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}}(9-x^2-y^2)\,dy\,dx$

c) $\int_{-3}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}}(x^2+y^2)\,dy\,dx$

d) $\int_{0}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}}(9-x^2-y^2)\,dy\,dx$

e) $\int_{-3}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}}(9-x^2-y^2)\,dy\,dx$

6. Which of the following options gives the value of the integral

$$\int_0^3 \int_{y^2}^9 y\cos(x^2)\,dx\,dy\ ?$$

a) $\frac{1}{2}\sin(81)$ b) $\frac{1}{4}\sin(81)$ c) $-\frac{1}{2}\sin(81)$

d) $\frac{1}{4}\cos(9)$ e) $-\frac{1}{4}\sin(81)$

7. Which of the following options gives the value of the integral

$$\int_0^1 \int_{\sqrt{x}}^1 ye^{y^2}\,dy\,dx\ ?$$

a) $e-2$ b) $\frac{e-2}{2}$ c) $\frac{e-1}{2}$ d) 1 e) $\frac{1}{2}$

8. Let a be a positive real number, and the function $f(x,y)$ is defined as follows:

$$f(x,y)=\begin{cases}\dfrac{\arctan(a(x^2+y^2))}{x^2+y^2}\cdot\arccos\left(\dfrac{x^3y}{x^4+y^2}\right), & (x,y)\neq(0,0)\\ 3\pi, & (x,y)=(0,0)\end{cases}$$

Since this function is continuous for every $(x,y)\in\mathbb{R}^2$, what is the value of a?

a) 2 b) 4 c) 3 d) 6 e) 5

9. Let c be a real number. For the function $z=x^c\cos\left(\frac{y}{x}\right)$ to satisfy the equation

$$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=2z$$

which of the following options gives the number c?

a) 0 b) -1 c) 3 d) 1 e) 2

10. Let D be the region bounded by the lines $y=x$, $y=6-x$, and the y-axis.

What is the volume of the solid that has a base as the region D bounded above by the plane $z=2x$ and below by the plane $z=0$?

a) 18 b) 9 c) 12 d) 24 e) 36

11. Let $D = \{(x, y) \in \mathbb{R}^2 : -\sqrt{3} \leq x \leq \sqrt{3},\ 0 \leq y \leq 2\}$. Which of the following is the value of the integral

$$\iint_D \frac{x^3 \cos(y) + 3}{x^2 + 1}\, dA \ ?$$

a) π b) 6π c) 4π d) 2π e) 3π

12. Let $z = f(x, y)$ be a differentiable function and

$$x = w \ln t + t \sin(\pi w), \quad y = te^{w-1} + w \sin\left(\frac{\pi t}{2}\right).$$

If

$$\left.\frac{\partial z}{\partial x}\right|_{(0,2)} = \frac{3}{\pi} \quad \text{and} \quad \left.\frac{\partial z}{\partial y}\right|_{(0,2)} = 4,$$

which of the following is the value of the partial derivative

$$\left.\frac{\partial z}{\partial w}\right|_{(w,t)=(1,1)} \ ?$$

a) 2 b) 4 c) 5 d) 3 e) 6

13. Which of the following options is the angle made with the x-axis by the direction in which the directional derivative of the function $f(x, y) = x^2 + y^2 - 2x - 4y$ at the point $(2, 3)$ is maximum?

a) $\frac{\pi}{3}$ b) 0 c) $\frac{\pi}{6}$ d) $\frac{\pi}{4}$ e) $\frac{\pi}{2}$

14.

$$f(x, y) = \begin{cases} \dfrac{x^2 - y^2}{3(x^2 + y^2)}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0) \end{cases}$$

Which of the following is or are true for the function above?

I. It is defined on the entire $\mathbb{R}^2$ plane.

II. Its limit at the point $(0, 0)$ is 0.

III. It is continuous at the point $(0, 0)$.

IV. The limit is 0 when approaching the point $(0, 0)$ along the line $y = x$.

a) II and III b) I only c) I, II, and IV
d) All of them e) I and IV

15. Which of the following is the double integral that gives the area of the triangle with vertices at $(0, 4)$, $(2, 0)$, and $(6, 4)$?

a) $\displaystyle\int_0^4 \int_{\frac{4-y}{2}}^{y+2} dx\, dy$

b) $\displaystyle\int_0^6 \int_{\frac{4-y}{2}}^{y+2} dx\, dy$

c) $\displaystyle\int_0^4 \int_{y+2}^{\frac{4-y}{2}} dx\, dy$

d) $\displaystyle\int_0^4 \int_{\frac{y-4}{2}}^{y+2} dx\, dy$

e) $\displaystyle\int_0^4 \int_{\frac{4-y}{2}}^{2-y} dx\, dy$

16. Let the function

$$f(x,y) = \sqrt{\ln\left(e^{x^2}y\right)}$$

be given. Which of the following equations is the equation of the tangent plane to the function at the point $(x,y,z) = (1,1,1)$?

a) $x+y-z=1$ b) $2x+y-2z=1$ c) $2x+y-z=2$
d) $x+2y-2z=1$ e) $2x-y-2z=1$

17. Let $f(x,y) = e^{y^2\tan y} + x^3y^3 + \ln(\cos x)$. Which of the following is the value of the expression

$$\left.\frac{\partial^6 f}{\partial x^3 \partial y^3}\right|_{(1,2)} ?$$

a) 0 b) 12 c) 18 d) 72 e) 36

18. Let

$$f(x,y) = \begin{cases} (x^3+y)\sin\left(\dfrac{1}{x^2+y^2}\right), & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0) \end{cases}$$

Which of the following gives the values of the partial derivatives $f_x(0,0)$ and $f_y(0,0)$ (if exist), respectively?

a) Both do not exist
b) 1 and does not exist
c) 0 and 0
d) 0 and does not exist
e) 1 and 0

19. Let $g(w,t)$ be a function with continuous partial derivatives. The equation

$$g(x-z, y-z^2) = 0$$

implicitly defines a function $z = f(x,y)$. Accordingly, which of the following is equal to the expression

$$\frac{\partial z}{\partial x} + 2z\frac{\partial z}{\partial y} ?$$

a) z b) $x+y$ c) $2z$ d) 0 e) 1

20. Which of the following is the largest domain of the function

$$f(x,y) = \frac{\sqrt{4-x^2-y^2}}{\ln(1+y-|x|)} ?$$

a) $\{(x,y) \mid x^2+y^2 \leq 4,\ y > |x|-1,\ y \neq |x|\}$
b) $\{(x,y) \mid x^2+y^2 < 4,\ y > |x|-1,\ y \neq |x|\}$
c) $\{(x,y) \mid x^2+y^2 \leq 4,\ y > |x|-1\}$
d) $\{(x,y) \mid x^2+y^2 < 4,\ y > |x|-1\}$
e) $\{(x,y) \mid x^2+y^2 \leq 4,\ y \geq |x|-1,\ y \neq |x|\}$